1. A retail chain wants to forecast product sales using factors like holidays, promotions, location, and inventory. The dataset has 80 features. How should they select the most impactful ones?

Feature & Model Selection - RFE, Regression

Step 1: Collect retail dataset which contains details like inventory, holidays, promotions, location
Step 2: Preprocess the data handling null values & encode categorical data
Step 3: use feature selection method like RFE or PCA to extract important features or columns that can be used for ML model building
Step 4: Create a ML model using linear regression or random forest or SVM to forecast the product sales
Step 5: Evaluate the model using RSqaure, SSR, SSE
Step 6: Deploy the model using pickle to see results

2.A podcast platform wants to recommend new podcasts to users based on the categories and hosts they follow. How should they design this content-based system?

Model type: Content based recommendation system

Step 1: Create content based interaction matrix from existing podcasts
Step 2: Calculate similarity between content (content based collaborative filtering) using cosine similarity
Step 3: Recommend podcasts that are similar to existing liked ones which target users haven't read
Step 4: Evaluate using RMSE or Recall
Step 5: Retrain the model for every podcasts added

3.An industrial firm wants to predict machine failure using sensor readings and system logs (200+ features). How do they choose which inputs to use?

Feature & Model Selection - RFE, Classification

Step 1: Collect machine dataset which contains details like sensory readings, system logs etc.
Step 2: Preprocess the data handling null values & encode categorical data
Step 3: use feature selection method like RFE or PCA to extract important features or columns that can be used for ML model building
Step 4: Create a ML model using logistic regression or random forest or SVM or decision tree to predict the machine failure
Step 5: Evaluate the model using Accuracy, precision, f1, ROC-AUC
Step 6: Deploy the model using pickle to see results

4. A fashion app wants to suggest outfits to users based on their previous likes, color preferences, and seasonal trends. How should they implement the recommendation?

Model type: User based Recommendation system

Step 1: Create user based interaction matrix from existing user preferences
Step 2: Calculate similarity between users (user based collaborative filtering) using cosine similarity
Step 3: Recommend outfits that are similar to previous likes, color preference & seasonal trends of the target users
Step 4: Evaluate using RMSE or Recall
Step 5: Retrain the model for every podcasts added

5.A research lab wants to predict heart disease using patient data. The dataset has lab results, habits, and vitals. How should they select only the most relevant features?

Feature & Model Selection - PCA, Classification

Step 1: Collect patient dataset which contains details like lab results, habits & vitals etc.
Step 2: Preprocess the data handling null values & encode categorical data
Step 3: use feature selection method like PCA or LDA to extract important features or columns that can be used for ML model building
Step 4: Create a ML model using logistic regression or random forest or SVM or decision tree to predict the heart disease
Step 5: Evaluate the model using Accuracy, precision, f1, ROC-AUC
Step 6: Deploy the model using pickle to see results

6. A job portal wants to recommend job listings based on a user's resume and previous applications. How can they build a recommendation system?

Model type: User based recommendation system

Step 1: Create user based interaction matrix from existing user resume and previous applications
Step 2: Calculate similarity between users (user based collaborative filtering) using cosine similarity
Step 3: Recommend jobs that are similar to previous applied, skill sets in the resume
Step 4: Evaluate using RMSE or Recall
Step 5: Retrain the model for every jobs added

7. A smart home system wants to predict energy usage using readings from multiple sensors. With 100+ features, how do they reduce complexity?

Feature & Model Selection - RFE, Regression

Step 1: Collect energy usage dataset which contains the reading from multiple sensors
Step 2: Preprocess the data handling null values & encode categorical data
Step 3: use feature selection method like RFE or PCA to extract important features or columns that can be used for ML model building
Step 4: Create a ML model using linear regression or random forest or SVM to forecast the energy usage
Step 5: Evaluate the model using RSqaure, SSR, SSE
Step 6: Deploy the model using pickle to see results

8. An e-learning platform wants to suggest tutorials based on a learner's skill gaps identified from quizzes. How should they build this system?

Model type: User based recommendation system

Step 1: Create user based interaction matrix from existing learners skills
Step 2: Calculate similarity between users (user based collaborative filtering) using cosine similarity
Step 3: Recommend courses that are required to the users based on previous learning, new courses
Step 4: Evaluate using RMSE or Recall
Step 5: Retrain the model for every courses added

9.A social media app wants to predict post engagement (likes/comments) using post features, timing, and user activity. How do they select the right features?

Feature & Model Selection - SelectKBest, Classification

Step 1: Collect post engagement dataset which contains details like features, timing, user activity etc
Step 2: Preprocess the data handling null values & encode categorical data
Step 3: use feature selection method like SelectKBest to extract important features or columns that can be used for ML model building
Step 4: Create a ML model using logistic regression or random forest or SVM or decision tree to predict the social media response
Step 5: Evaluate the model using Accuracy, precision, f1, ROC-AUC
Step 6: Deploy the model using pickle to see results

10. Now you want to create an app named `attendance` in the project and register it.write logic for this Django project

Step 1: Open terminal & check if you’re in the right folder, if not use **CD** to change the folder

Step 2: Run python manage.py startup attendance

Step 3: Open **`student_portal/settings.py`.**

Step 4: In `INSTALLED_APPS`, add this line: **attendance**